

Medify

Smart Medication Management System

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The Problem

Why medication management is a critical challenge



- Patients taking multiple medications face high risk of missed or incorrect intake
- Passive solutions (pill organizers) provide no reminders or confirmation
- Complex apps are not suitable for elderly or cognitively impaired users
- Caregivers and family members have no visibility into daily intake compliance



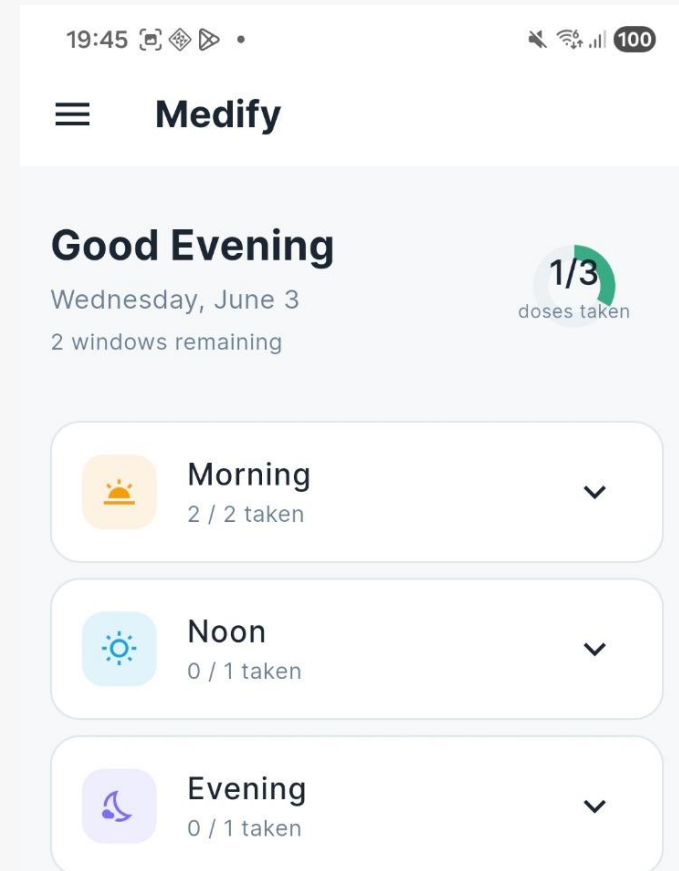
Our Solution

Bridging the gap between prescriptions and actual intake

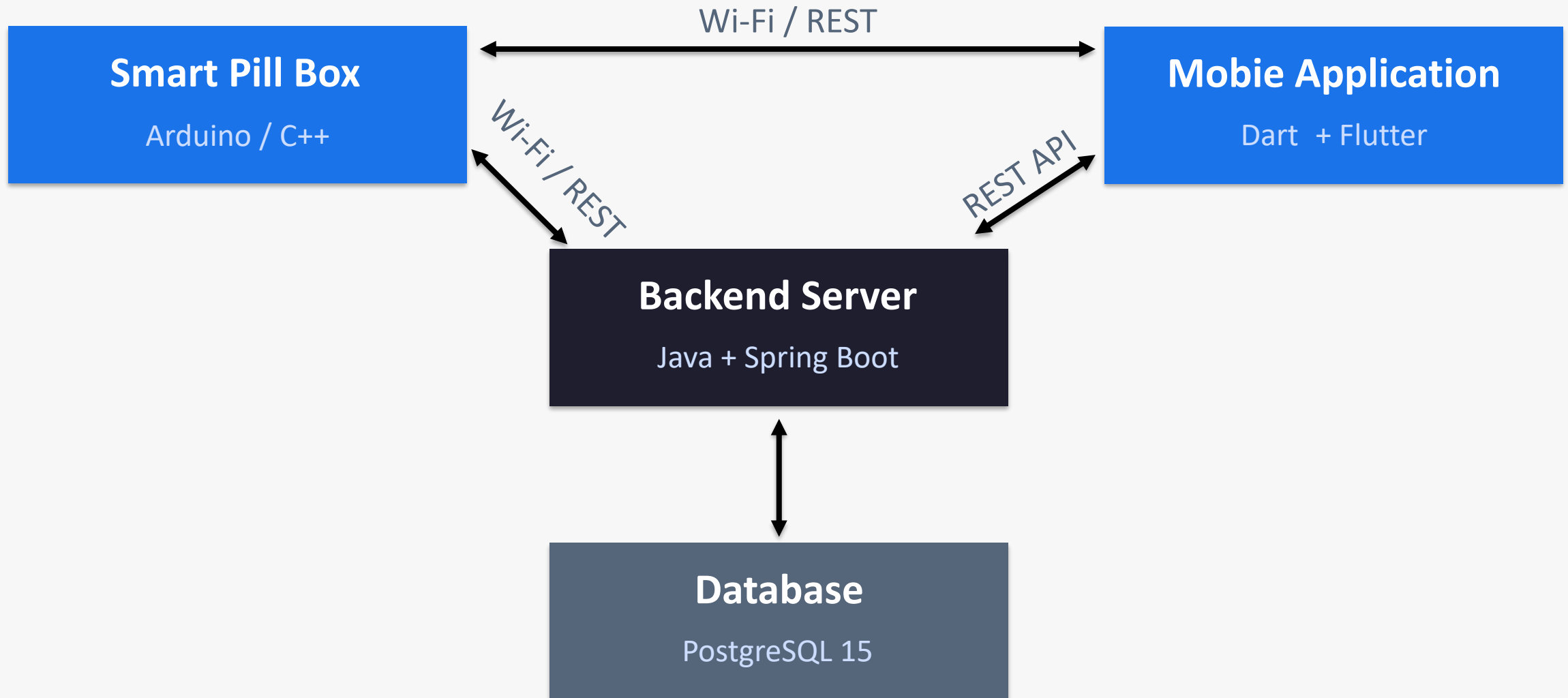


From passive pill box to active medication management

Medify is a smart medication management system that combines a smart pill box and a mobile app. The system actively guides and monitors the medication intake process, from medication management, reminders and caregiver alerts to medication dispensing.



System Architecture



Technology Stack

Choices and justifications



Backend

Java + Spring Boot

Stable backend platform with built-in support for REST APIs, scheduling, and database integration.

Frontend

Dart + Flutter

Single codebase for Android and iOS with fast performance and responsive UI.

Hardware

Arduino / C++

Simple and accessible platform for embedded systems, with extensive hardware library support and object-oriented development in C++.

Database

PostgreSQL

Reliable relational database with a fixed schema and strong Spring Boot integration.

Domain — Mobile Application

Flutter + Dart



Medication Management

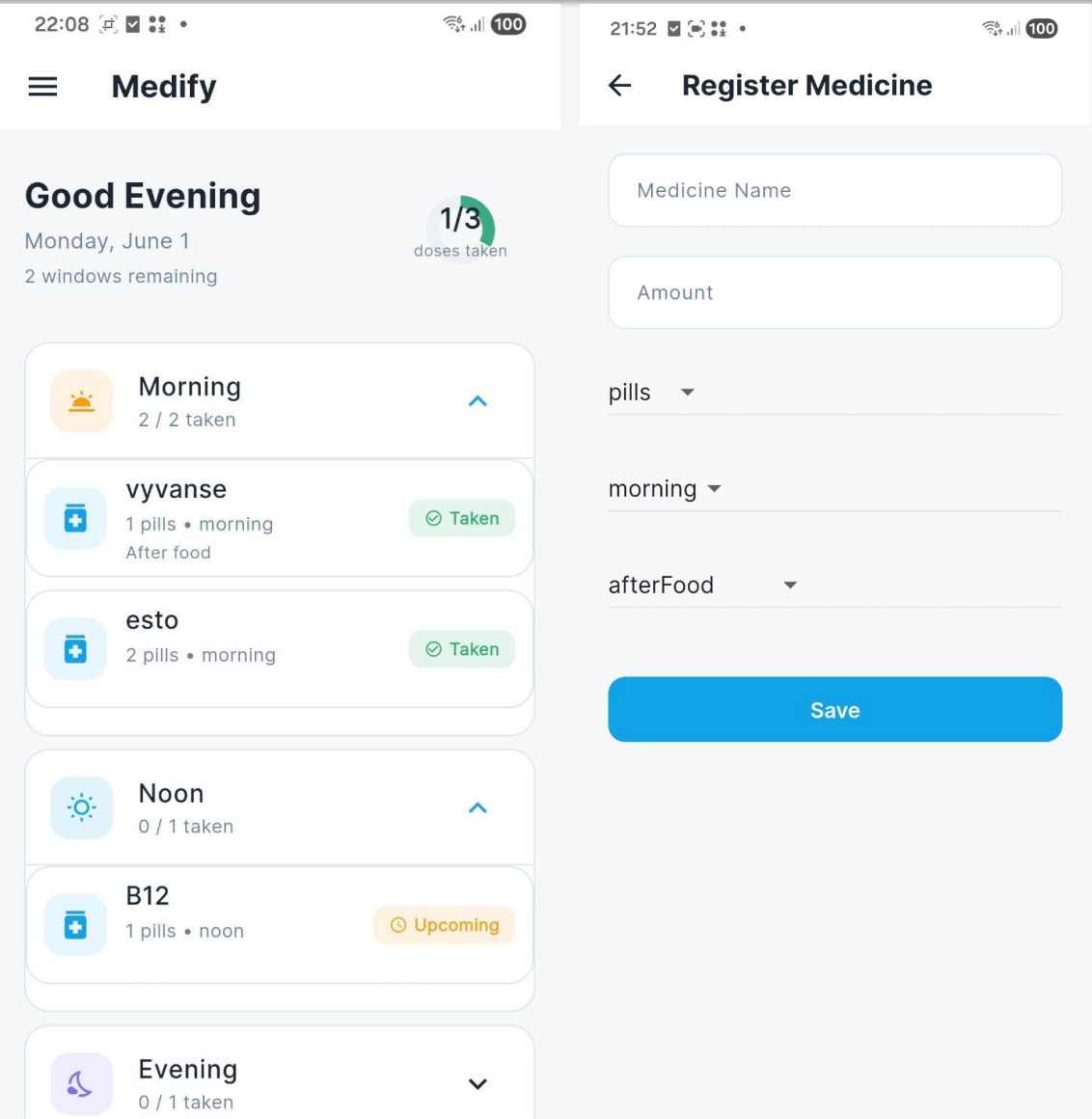
Manage medications and treatment schedules through the app.

Accessible Experience

Simple UI designed to support elderly users.

Medication Setup

Guides medication loading into the smart pill box.



Domain — Mobile Application

Flutter + Dart

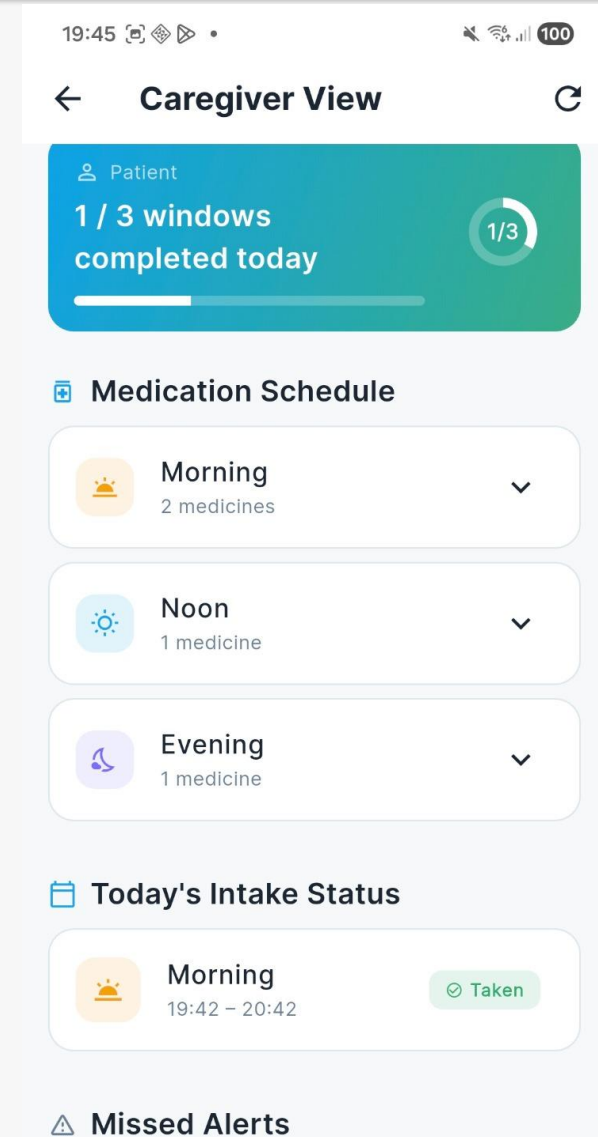
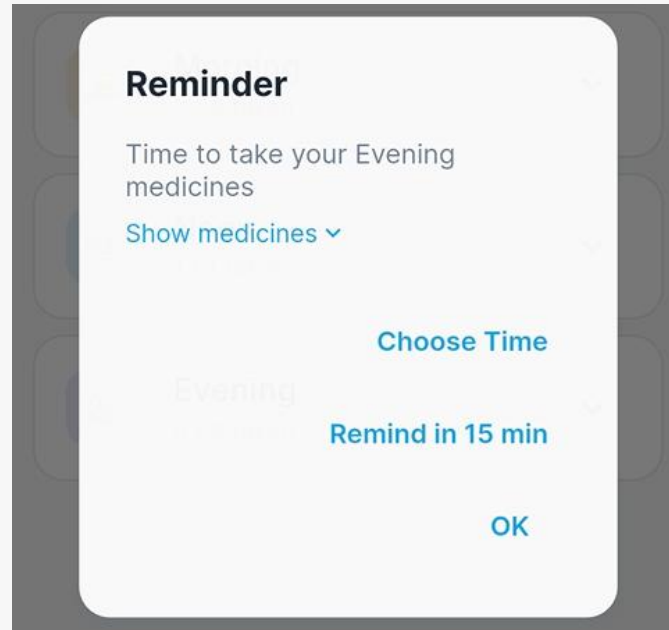


Notifications & Reminders

Medication reminders with approval and snooze options.

Caregiver View

Real-time monitoring and alerts for caregivers.



Domain — Smart Pill Box

Arduino / C++



Medication Release

Motor-driven mechanism releases medication after user approval.

Compartment Monitoring

IR sensor detects whether the intake compartment was emptied.

Autonomous Operation

Operates independently even when the patient's phone is not nearby.



Domain — Backend Server

Java 21 · Spring Boot 3.4



Scheduling

Handles intake-related events based on predefined schedules.

Intake Management

Manage the intake status lifecycle: PENDING → APPROVED → TAKEN.

Data Management

Manages application data and database operations.

Missed intake detection

Detects missed medication intake and sends alerts to caregivers.

Modules Structure

```
medify-backend
├── .claude
├── .idea
├── api
├── app
├── data
├── domain
├── .gitignore
├── CLAUDE.md
├── docker-compose.yml
├── pom.xml
└── README.md
```

- app - Entry point.
- api - REST controllers, services.
- Domain - Business logic, schedulers, models.
- data - DB communication and access.

Domain — Database

PostgreSQL



users

User profiles, roles, and authentication.

medications

Medication details and schedules.

intakes

Intake history and status tracking.

caregiver_links

Patient-caregiver connections and alert preferences.

notification_logs

System's notification and alert history.

devices

Registered pill box devices, IP addresses and status.

Current Project Status



Completed

- ✓ Patient app - register medications, intake schedule, notifications with approve or snooze.
- ✓ Caregiver screen - intake schedule and daily status overview.
- ✓ Backend core - automatic scheduling, intake tracking, notifications and persistent DB storage.
- ✓ Pill Box - Wi-Fi communication, API handling, and app-controlled medication dispensing.

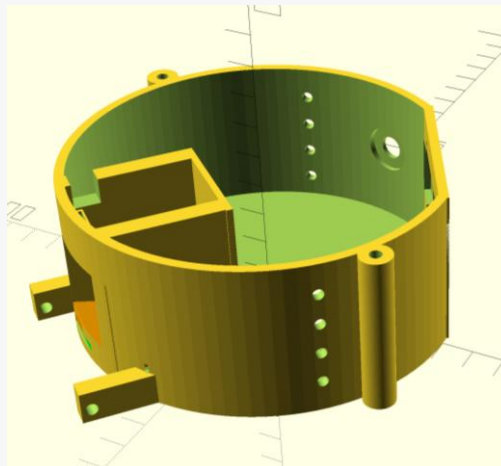
Planned

- Medication intake detection using IR sensors.
- Medication loading workflow through the mobile application.
- Multi-user and multi-caregiver support.

AI Usage in Development



- **Lovable** - used for UI planning, screen prototyping, and improving screen designs before implementing them in Flutter.
- **Claude Code** - used for code generation, architecture discussions, debugging assistance, and project documentation.
- **ChatGPT** - used for document phrasing, hardware design discussions, and assistance with OpenSCAD based 3D printed components.

A screenshot of the Medify mobile application interface. At the top, there's a header with the Medify logo and a hamburger menu icon. Below the header, it says "Good Morning" followed by the date "Wednesday, June 3" and "All done for today". To the right of this is a circular pill counter showing "0/0 doses taken". The main content area is titled "Manage your medications" and includes a "Back" button. There are four input fields: "Medication Name" (with placeholder "Enter medication name"), "Dosage" (with "Amount" and "Pills" dropdowns), "Time of Intake" (with "Morning" dropdown), and "Instructions" (with "On an empty stomach" dropdown). At the bottom is a green button with a plus sign and the text "Add Medicine".

System Demo

See Medify in action



Thank you!

